

Answers

VARC

1. A	2. B	3. D	4. C	5. D	6. B	7. C	8. B
9. C	10. D	11. C	12. C	13. C	14. C	15. D	16. A
17. B	18. D	19. 2	20. 3	21. 4123	22. 4123	23. C	24. C

LRDI

25. 0	26. 200	27. D	28. B	29. A	30. 3	31. C	32. D
33. 21	34. A	35. 0	36. 0	37. 3	38. 4	39. C	40. A
41. D	42. D	43. B	44. B				

Quant

45. B	46. C	47. C	48. B	49. 3	50. 2	51. 6	52. C
53. D	54. C	55. C	56. A	57. C	58. 972	59. 20808	60. 27
61. B	62. C	63. 2	64. B	65. C	66. 19		

Explanations

VARC

1. **A**

The passage focuses on the interconnectedness within the global south in the context of the Indian Ocean world's migration networks. It emphasizes historical connections between the coasts of East Africa, the Arab coasts, and South and East Asia. The passage does not specifically highlight migration networks connecting the Indian Ocean world with the global north. Instead, it underscores the significance of geographical location, religious histories, and commercial interactions within the region, pointing to a more localized and regional perspective on migration. Therefore, **Option A** is not true according to the passage. Additionally, the passage mentioned the Indian Ocean as "a term used to describe the very long-lasting connections among **the coasts of East Africa, the Arab coasts, and South and East Asia.**" and not north and south.

Option B is correct as the passage mentions that for much of history, travel by sea in the Indian Ocean was easier than by land, emphasizing the importance of geographical location.

Option C is correct as the passage indicates that the novels in the book draw on and shape a wider sense of Indian Ocean space through themes, images, metaphors, and language, including religious and commercial aspects.

Option D is correct as the passage notes that migration is often portrayed as abandonment rather than adventure, indicating a complex and ambivalent nature of the migration experience in the Indian Ocean world.

[VIDEO SOLUTION](#)

2. B

Options A, C and D have the following format World/Novels : Characteristic of that particular world/novel. Option B is the odd one out as the characteristic of Border-crossing does not belong to the Postcolonial novels world.

From the passage, we can infer that the Indian Ocean novels were "outward-looking - full of movement, border-crossing and south-south interconnection". At the same time, they showcased elements of the global south like Slavery, Forced Migration etc. Hence, A and C showcase valid elements of the Indian Ocean Novels World.

On the other hand, postcolonial novels were usually [concerned with] national questions, land-focused and inward-looking. They featured anti-colonial nationalism. Hence, option D is also a valid Theme:Characteristic combination.

However, we note that Border-crossing is an element of the Indian Ocean novel world and not the Postcolonial novel world. Hence, option B is not a valid combination and thus is the odd one out.

VIDEO SOLUTION

3. D

"For their part Ghosh, Gurnah, Collen and even Conrad reference a different set of histories and geographies than the ones most commonly found in fiction in English. Those [commonly found ones] are mostly centred in Europe or the US, assume a background of Christianity and whiteness, and mention places like Paris and New York."

The passage argues that the novels discussed in "Writing Ocean Worlds" diverge from the common representations found in English fiction, which often center on Europe or the US, assume a background of Christianity and whiteness, and mention places like Paris and New York. If **Option D** were true, it would support the passage's claim rather than weaken it. Therefore, Option D is the correct answer.

Through the passage, the author claims that the Indian Ocean novels provide a more realistic picture of the Indian Ocean space, particularly in the representation of Africa. The author claims that the depiction is more authentic and free from Eurocentricity that is seen in other novels.

Option A weakens the passage by contradicting these claims and suggesting that the depiction of Africa is influenced by postcolonial nostalgia.

Option B weakens the passage by suggesting a potential bias or negative stereotyping in the portrayal of Africa in Indian Ocean novels.

Option C weakens the author's claim by disputing that there is eurocentric perspective in other novels.

VIDEO SOLUTION

4. C

Option C is the correct answer because it contradicts the idea of "remapping" discussed in the passage. The passage emphasizes that the novels under consideration challenge the common representations found in English fiction, particularly those centered in the West. Option C, suggesting that cosmopolitanism originated in the West and traveled to the East through globalization, aligns with the conventional Western-centric narrative rather than the passage's argument of reshaping perspectives and centralizing the interconnected global south, particularly the Indian Ocean world, as a key space in the reimagined literary landscape.

Option A aligns with the passage's discussion of the novels focusing on the Indian Ocean world, contributing to the "remapping" beyond national concerns.

Option B aligns with the passage's emphasis on the interconnected Indian Ocean world, challenging the Eurocentric perspective on trade and commerce.

Option D supports the passage's claim that historical evidence suggests that globalization first appeared in the Indian Ocean, contributing to the "remapping" of the world's historical and geographical perspectives.

VIDEO SOLUTION

5. D

Option D is not explicitly presented by the author as a reason why non-geographers disregard geographic influences. The author suggests that scholars often react negatively to explanations involving a geographic role by denouncing "geographic determinism." However, the specific idea of dismissal is not explicitly outlined in the passage.

The other options on the other hand, can be inferred from the passage:

Option A can be inferred from the following lines: *"One reason is that some geographic explanations advanced a century ago were racist, thereby causing all geographic explanations to become tainted."*

Option B can be inferred from the following lines: *"Another reason for reflex rejection of geographic explanations is that historians have a tradition, in their discipline, of stressing the role of contingency (a favorite word among historians) based on individual decisions and chance."*

Option C can be inferred from the last paragraph of the passage: *"Geographic explanations usually depend on detailed technical facts of geography and other fields of scholarship . . . Most historians and economists don't acquire that detailed knowledge as part of the professional training."*

VIDEO SOLUTION



6. B

The passage does not explicitly mention the criticism of scholars for having outdated interpretations of past cultural and historical phenomena. The primary focus of the author's criticism, as discussed in the passage, centers on scholars' tendencies to dismiss geographic factors, label geographic explanations as deterministic, and associate geographic analyses with past racism. Therefore Option B is the correct answer.

VIDEO SOLUTION

7. C

Option C cannot be directly inferred from the passage. The passage does discuss the influence of both geographic factors (such as biogeography) and non-geographic factors (culture, history, individual decisions) on human phenomena. However, the passage does not explicitly quantify or compare the prevalence of these influences by stating that *"most human phenomena result from culture and individual choice."*

Option A: The author discusses the development of warm fur clothes among the Inuit living north of the Arctic Circle and asserts that it was not due to a specific individual decision or historical contingency in 1783 but rather a response to environmental factors.

Option B: The author discusses the current state of agricultural practices in Australia, stating that the crops and domestic animals that make Australia a food and wool exporter today are all non-native species (mainly Eurasian) brought to Australia by overseas colonists. The use of the term "non-native" implies a change in agricultural practices from what was originally present in the Australian continent.

Option D: The author mentions that some geographic explanations advanced a century ago were racist, causing all geographic explanations to become tainted by racist associations in the minds of many scholars.

VIDEO SOLUTION

8. B

Option B is the correct answer because the passage uses examples like the Inuit and Aboriginal Australians to illustrate the influence of physical circumstances, particularly environmental factors, on human behavior and cultural practices. The discussion about the development of warm fur clothes among the Inuit due to the Arctic environment and the absence of indigenous farming in Aboriginal Australia because of the lack of domesticable native species underscores how physical circumstances dictate certain aspects of human behavior and shape cultural adaptations. Therefore, Option B accurately captures the main idea conveyed by the examples provided in the passage.

Option A is not explicitly emphasized in the passage; the focus is more on how environmental factors influence behavior and cultures.

Option C: The passage doesn't explicitly highlight self-sufficiency but rather the impact of specific environmental factors on the development of societies.

Option D is not entirely incorrect, but Option B more precisely captures the emphasis on physical circumstances dictating human behavior and cultures in the context of the examples provided in the passage.

VIDEO SOLUTION

9. C

The passage mentions that as early as the ninth century, the royal office of the Luparii, or wolf-catchers, was created in France to tackle the predators. However, this office became redundant as it had finished its job (kill the last wolf). So the resurgence of the wolves can't be attributed to it shutting down. The other options on the other hand, can be clearly inferred.

Option A: "Various factors explain the changes of the past few decades. Rural depopulation is part of the story. In Lozère, for example, farming and a once-flourishing mining industry supported a population of over 140,000 residents in the mid-19th century. Today the department has fewer than 80,000 people, many in its towns."

Option B: "As humans withdraw, forests are expanding. In France, between 1990 and 2015, forest cover increased by an average of 102,000 hectares each year, as more fields were given over to trees. Now, nearly one-third of mainland France is covered by woodland of some sort."

Option D: "The mostly protected status of the wolf in Europe—hunting them is now forbidden, other than when occasional culls are sanctioned by the state—plus the efforts of NGOs to track and count the animals, also contribute to the recovery of wolf populations."

VIDEO SOLUTION

10. D

Considering the first paragraph: "RESIDENTS of Lozère, a hilly department in southern France, recite complaints familiar to many rural corners of Europe. In remote hamlets and villages, with names such as Le Bacon and Le Bacon Vieux, **mayors grumble about a lack of local schools, jobs, or phone and internet connections.** Farmers of grazing animals add another concern: the return of wolves. Eradicated from France last century, the predators are gradually creeping back to more forests and hillsides. "The wolf must be taken in hand," said an aspiring parliamentarian, Francis Palombi, when pressed by voters in an election campaign early this summer. Tourists enjoy visiting a wolf park in Lozère, **but farmers fret over their livestock and their livelihoods.**"

Options A, B and C can be clearly inferred from the highlighted part.

The passage mentions that the number of people holding hunting licenses is still high but the number of people who still actively hunt is low. So Option D which states that there is decline in the number of hunting licences is incorrect.

VIDEO SOLUTION

CAT Percentile Predictor



11. C

The author's claims seem to be focused on the conflicts between farmers and the return of wolves, the economic implications, and the coexistence challenges. If wolf attacks on tourists were on the rise, it might shift the narrative and suggest a broader safety concern beyond the impact on farmers, potentially weakening the author's emphasis on the positive economic aspects of wolf-related tourism. Therefore Option C, if true, would weaken the author's argument.

Option A supports the author's claims about the return of wolves to Lozère.

Option B is not directly related to the author's claims about conflicts between farmers and wolves or the economic implications of wolf-related tourism.

Option D, if true, would not necessarily weaken the author's claims but might be seen as providing additional information about land use in Lozère.

VIDEO SOLUTION

12. C

"As the lupine population of Europe spreads westwards, with occasional reports of wolves seen closer to urban areas, expect to hear of more clashes between farmers and those who celebrate the predators' return. Farmers' losses are real, but are not the only economic story. Tourist venues, such as parks where wolves are kept and the animals' spread is discussed, also generate income and jobs in rural areas."

The passage mentions that farmers in Lozère are concerned about the return of wolves causing losses in livestock. On the other hand, environmentalists view the presence of wolves as a sign of wider ecological health. The suggested economic solution involves tourist venues related to wolves, such as parks, which not only address the economic concerns of farmers by generating income but also align with the interests of environmentalists who appreciate the return of the predators. Therefore, Option C accurately captures the collaboration between farmers and environmentalists in the proposed solution.

VIDEO SOLUTION

13. C

The passage emphasizes that Marshall Sahlins's main goal in writing his essay was to hold a mirror to an acquisitive society (contemporary economic life and bourgeois individualism). The essay accomplishes this by providing examples of foraging societies that made real choices to prioritize values such as leisure, mobility, and freedom over material accumulation. Sahlins contrasts the Zen road to affluence, where affluence is achieved by wanting less, with the capitalist pursuit of wealth through material production and consumerism. Therefore, Sahlins's goal, as portrayed in the passage, aligns with the idea of presenting examples of communities that have successfully chosen non-materialistic paths as a critique of acquisitive societies. So, Option C is the correct answer.

Option A: While Sahlins's essay acknowledges growing inequality and social hierarchies resulting from the Neolithic Revolution, it is more focused on contrasting foraging societies with contemporary economic life.

Option B: The primary emphasis is on showcasing foraging societies' choices and values rather than asserting a progressive degeneration of society.

Option D: Even though Sahlins's essay critiques aspects of contemporary economic views, its primary focus is not explicitly countering Galbraith's pessimistic view but rather presenting alternative possibilities through examples of non-materialistic societies.

VIDEO SOLUTION

14. C

Option C is the correct answer because the passage uses the example of Tanzania's Hadza community to illustrate that forager communities, like the Hadza, do not conform to a simple narrative of human diversity or victimhood. Instead, they actively make choices about their way of life. The passage mentions that the Hadza, despite being surrounded by farmers, knew they had alternatives and consciously rejected them. This example serves to emphasize that forager communities are not constrained by ignorance about alternatives; rather, their lifestyles derive from their own choices. Therefore, Option C accurately captures the essence of the Hadza illustration in the passage.

Option A: The passage doesn't specifically highlight the survival techniques of hunter-gatherer communities into contemporary times, but rather emphasizes their choices and values.

Option B: The passage doesn't discuss the Hadza community in the context of agrarian practices in contiguous communities, making this option irrelevant to the illustration.

Option D: The passage does not suggest that the Hadza community coexisted with vastly different ways of living and working for centuries.

VIDEO SOLUTION

15. D

The passage explicitly mentions that Sahlins's essay, "The Original Affluent Society," brought a critical perspective to contemporary consumerism and inequality, echoing the themes found in John Kenneth Galbraith's work, "The Affluent Society." The passage notes that Sahlins's essay contrasts the values of foraging societies with the capitalist pursuit of wealth, and it suggests that the essay complements Galbraith's skeptical portrait of postwar prosperity and inequality. Therefore, Option D accurately reflects the information presented in the passage regarding the relationship between Sahlins's views and Galbraith's criticism of contemporary society.

The passage does not suggest that Galbraith's theories refute Sahlins's thesis but rather highlights their complementarity (Option A) nor does it focus on contrasting foragers' ways of living with Galbraith's views on contemporary growth paths (Option B).

The passage does not document the influence of Galbraith's views on Sahlins's analysis; instead, it emphasizes how Sahlins's essay complements Galbraith's critical perspective on contemporary society. Therefore Option C is incorrect too.

VIDEO SOLUTION



16. A

"Viewed in today's context, of course, not every aspect of the essay has aged well. While acknowledging the violence of colonialism, racism, and dispossession, it does not thematize them as heavily as we might today."

Option A is the correct answer because the passage explicitly mentions that, when viewed in today's context, not every aspect of Sahlins's essay has aged well, and it acknowledges that the essay does not thematize issues like racism, colonialism, and dispossession as heavily as might be expected today. The term " cursory treatment " suggests that the essay provides only a brief or superficial examination of the effects of racism and colonialism on societies, and the passage criticizes this aspect of the essay for not giving these important issues more comprehensive attention.

VIDEO SOLUTION

17. B

The sentence best fits in Blank 3 because it logically follows the mention of the unexpected genetic link between Native Americans and Japanese people mentioned in the sentence preceding Blank 3. After establishing this genetic connection, the sentence provides additional context by explaining the archaeological implications of the discovery. It suggests that the shared genetic link has archaeological manifestations, leading to similarities in the Paleolithic peoples of China, Japan, and the Americas. Placing the sentence here helps to connect the genetic findings to broader archaeological and cultural aspects.

Blank 1: The sentence before Blank 1 introduces the unexpected genetic link between Native Americans and Japanese people. Placing this sentence in Blank 1 would disrupt the logical progression of information, as Blank 1 should provide information that directly connects with or follows from the mention of the unexpected genetic link, and the sentence is more relevant to explaining the broader context after the genetic link has been introduced.

Blank 2: Putting the sentence here makes no sense as there has been no mention of any discovery before.

Blank 4: Placing the sentence here would disrupt the flow of the passage as before Blank 4 the passage mentions how there were similarities between them and After Blank 4, the passage states " This suggests.. ". " This " clearly refers to the similarities mentioned before Blank 4.

VIDEO SOLUTION

18. D

The sentence best fits in Blank 2 because it provides a continuation and elaboration on the disquieting experience mentioned before Blank 2. Before Blank 2, the passage introduces the disquieting nature of reading philosophy, and the sentence builds on that by explaining that the philosophical examination challenges the values around which one has organized their life, making them appear provincial, flatly wrong, or even evil. The sentence in question serves to articulate the difficulty and discomfort associated with this profound philosophical scrutiny, logically following the initial statement about the disquieting experience of reading philosophy.

We can also see that the sentence ends with "...and way of life is difficult enough." This is immediately followed by the passage highlighting what's worse "What's worse, philosophers admonish each other to remain unsutured until such time as a defensible new answer is revealed or constructed." So we can see that Sentence 2 serves as a preceding sentence to this.

VIDEO SOLUTION

19.2

Sentence 2 is the odd one out because it introduces a term, "theory of mind," and discusses scholars' preferences for alternative terms like "mentalizing" or "mindreading." Unlike the other sentences, which focus on explaining and elaborating on the concept of understanding others' mental states, sentence 2 provides more of a meta-discussion about the terminology used to describe this capacity rather than directly contributing to the explanation of the topic. The other sentences contribute to the substantive discussion of the capacity to perceive and interpret other people's behavior in terms of their mental states, making sentence 2 the odd one out in the context of forming a coherent paragraph.

VIDEO SOLUTION

20.3

Sentence 3 is the odd one out because it introduces a different topic compared to the other sentences. While the other sentences focus on the naming patterns of numbers in English, especially the irregularities and variations in the system, sentence 3 shifts the focus to the learning process of children and their understanding of the differences between numbers like "fourteen" and "forty." The other sentences contribute to the discussion about the intricacies of English number naming, making sentence 3 less aligned with the central theme of the paragraph.

VIDEO SOLUTION



21.4123

The correct order is 4-1-2-3.

Sentence 4 introduces the central question about the enduring fascination with certain crimes and the perpetual re-examination of some cases.

Sentence 1 builds on this question by specifically asking about the "unusual elements" that make a particular case attractive to a certain audience, providing a more focused inquiry.

Sentence 2 follows by suggesting possible reasons, such as a particularly savage or unfathomable level of depravity, and the role of mystery in drawing attention to these cases.

Sentence 3 then generalizes the idea, stating that unsolved and perhaps unsolvable cases offer something unique that "ordinary" murder cases don't, emphasizing the enduring allure of unresolved mysteries in the realm of crime.

VIDEO SOLUTION

22.4123

4-1-2-3 is the correct order.

Sentence 4 introduces the idea that biased decisions made by humans have a localized impact, but with AI's emergence, the impact scale becomes much broader. This sets the context for the discussion to be followed.

Now, if we consider Sentences 1 and 2 we can see that Sentence 1 is expanding on the impact of biased decisions in the context of AI, highlighting that algorithms hosted on the internet, accessed by many, result in more significant adverse effects on larger groups of people. Sentence 2 is providing a clarification on the term "algorithmic bias," emphasizing that the foundation of bias lies in the data rather than the algorithms themselves. It explains that algorithms reflect persistent patterns present in the training data. Therefore we can say that Sentence 2 must be following 1.

Sentence 3 is pointing out that despite the widespread impact of AI biases, it is comparatively easier to fix them than human-generated biases, as it is simpler to identify and address biases in algorithms than to make people unlearn behaviors learned over generations. This logically follows the point made in Sentence 2. Therefore Sentence 3 must be following Sentence 2.

Therefore the correct order is 4-1-2-3.

VIDEO SOLUTION

23.C

The passage discusses the historical use of misinformation for political purposes, dating back to ancient Rome with Octavian's smear campaign against Antony. It then highlights how the 21st century has seen an unprecedented scale of information weaponization, facilitated by powerful technology and amplified through social networks. Option C effectively conveys the continuity of using misinformation for power throughout history, now fueled by modern technology. Therefore, Option C is the correct answer.

Option A: While the passage acknowledges the historical aspect of disinformation, it emphasizes the unprecedented scale in the 21st century, which is not captured in this option.

Option B focuses on the need for critical reading without explicitly highlighting the historical context and the weaponization of information for power.

Option D does not emphasize the broader historical and contemporary context of misinformation for political purposes.

VIDEO SOLUTION

24.C

Option C is the correct answer because it accurately captures the main idea of the passage. It highlights how technological advancements in navigation during the sixteenth century transformed colonialism by enabling Europeans to establish settlements and exert political dominance over distant regions, including the Americas, Australia, and parts of Africa and Asia.

Option A focuses on the displacement of indigenous populations, which is not the central point of the passage.

While Option B mentions the expansion of territories and political power, it does not emphasize the technological advancements in navigation.

Option D introduces the concept of British settlements, which is narrower than the broader context of European colonialism discussed in the passage.

[VIDEO SOLUTION](#)

LRDI

Explanation [25 - 29]:

It is given that the applications are scheduled for processing in twenty 15-minute slots starting at 9:00 am and ending at 2:00 pm. Ten applications are scheduled in each slot.

Hence, the total number of applicants = $(20 \times 10) = 200$. It is also known that 50% of the applications were US applications, and the number of US applications was the same in all the slots. The same was true for the other three categories.

Hence, the number of total number of US applicants = $(200 \times 50\%) = 100$, and the number of US applicants in each slot = $(100/20) = 5$

It is also known that Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot. Since the number of Schengen applicants was the same in all the slots, it implies the number of Schengen applicants in each slot is at least 3.

Similarly, it is given that Mahira and Osman were scheduled in the 9:30 am slot on that day for visa processing in the Others category, which implies the number of other category applicants in each slot is at least 2. Since the number of total applicants in each slot is 10, this implies the number of Schengen and other applicants in each slot is 3, and 2, respectively. Hence, the number of UK applicants is 0 in each slot.

It is also known that the number of total counters is 10, among which four are dedicated to US applications, and two each for UK applications, Schengen applications, and Others applications. It is given that each US and UK application requires 10 minutes of processing time, and Vijay was called to a counter at 9:25 am. (Who is 5th in the queue). It can only be possible when the processing time of Schengen applications is 12.5 minutes.

US (10 min)				Schengen (12.5 min)		Others (5 min process)	
End Time				End Time		End Time	
C1	C2	C3	C4	C1	C2	C1	C2
9.10	9.10	9.10	9.10	9.12.30	9.12.30	9.05	9.05
9.20	9.25	9.25	9.25	9.25	9.32.30	9.20	9.20
9.30	9.35	9.40	9.40	9.37.30	9.45	9.35	9.35
9.40	9.45	9.50	9.55				
9.55	9.55	10.00	10.05				
10.10	10.10	10.10	10.15				
10.20							

On a particular day, Ira, Vijay, and Nandini were scheduled for Schengen visa processing in that order. They had a 9:15 am slot but entered the VPO at 9:20 am. When they entered the office, exactly six out of the ten counters were either processing applications, or had finished processing one and ready to start processing the next. Hence, at 9:20 am, there are exactly four free counters. Out of these 4, 2 is the UK counter, and the other two are other counters. (Since the US counters and Schengen Counters were either processing applications, or had finished processing one and were ready to start processing the next.)

25. 0

From the table, we can say that the total number of UK applicants in each slot is zero, Hence, the total number of applicants is zero.

[VIDEO SOLUTION](#)



26. 200

For the other applicants, the time taken to process one application is at most 5 minutes, which implies the total time taken to process 40 applications is at most $(40 \times 5) = 200$ minutes.

[VIDEO SOLUTION](#)

27. D

Nandini's position was sixth in the queue in the Schengen Applications. From the table, we can see that her process will end at 9.45 am.

The correct option is D

[VIDEO SOLUTION](#)

28. B

Let's check the options.

Option A: The application process of Osman was completed before 9:45 am. => True (Since he is 5th in the queue, his process will end at 9.35 am)

Option B: The application process of Mahira started after Nandini's. => The application process for Mahira starts at 9.30 am, and the application process for Nandini starts at 9.32.30 am => False.

The correct option is B

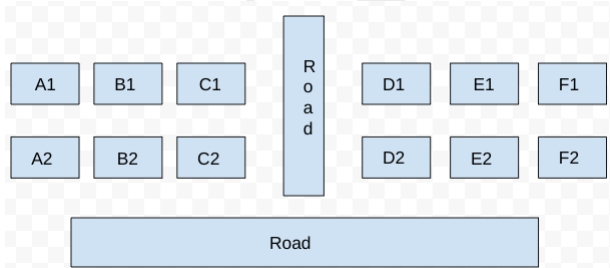
29. A

From the table, we can see that the first slot takes 20 minutes to complete, and after that remaining 19 slots take 15 minutes each to complete the US application process.

Hence, the total time taken = $20 + 15 \times 19 = 305$ minutes = 5 hrs 5 minutes. Hence, the time will be (9 am + 5hrs 5 minutes) = 2.05 pm

The correct option is A

Explanation [30 - 34]:



It is given that some of the houses are occupied. The remaining ones are vacant and are the only ones available for sale.

The base price of a vacant house is Rs. 10 lakhs if the house does not have a parking space, and Rs. 12 lakhs if it does. The quoted price (in lakhs of Rs.) of a vacant house is calculated as (base price) + $5 \times$ (road adjacency value) + $3 \times$ (neighbor count).

It is also known that the maximum quoted price of a house in Block XX is Rs. 24 lakhs

Hence, there can be two cases for the maximum quoted price of a house in block XX.

Case 1: House with parking space:

$$\Rightarrow 12 + 5a + 3b = 24 \Rightarrow 5a + 3b = 12 \quad (a = \text{road adjacency value}, b = \text{neighbor count})$$

The only value for which the equation satisfies is ($a = 0$, and $b = 4$). But the value of b can't be 4 because the maximum neighbor count can be at most 3.

Hence, case 1 is invalid.

Case 2: House without parking space:

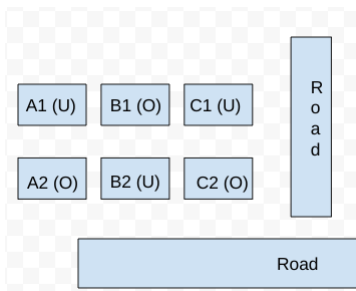
$$\Rightarrow 10 + 5a + 3b = 24 \Rightarrow 5a + 3b = 14$$

$$\Rightarrow (a, b) = (1, 3)$$

Hence, the house must have 3 neighbors and 1 road connected to it. Hence, the only possible case is B2. Therefore, the neighbor houses of B2, which are (B1, A2, and C2) are occupied.

It is known that Row 1 has two occupied houses, one in each block. Since B1 is already occupied, it implies A1, and C1 are vacant.

Hence, the configuration of block XX is given below: (Where U = Unoccupied/ Vacant, and O = Occupied)



Now for block YY, we know that both houses in Column E are vacant. Each of Column-D and Column-F has at least one occupied house. There is only one house with parking space in Block YY.

It is also known that the minimum quoted price of a house in block YY is Rs. 15 lakhs, and one such house is in Column E.

Case 1: The minimum quoted house is E2:

We know that the road adjacency of E2 is 1, hence we can calculate whether the house has parking space or not, and the neighbor count (b)

If the house has parking space, then: $12 + 5 \times 1 + 3 \times b = 15 \Rightarrow 3b = -2$ (which is not possible)

Hence, the house has no parking space $\Rightarrow 10 + 5 \times 1 + 3b = 15 \Rightarrow b = 0$

$b = 0$ implies all the neighbor house of E2 is vacant, which are (E1, D2, and F2).

It is known that each of Column-D and Column-F has at least one occupied house, which implies D1, and F1 must be occupied.

But D1 and F1 can't be occupied together since the total number of occupied houses in Row 1 is 2 (one in each block).

Hence, This case is invalid.

Case 2: The minimum quoted house is E1:

We know that the road adjacency of E1 is 0, hence we can calculate whether the house has parking space or not, and the neighbor count (b).

i) If the house has no working space, then: $10 + 5 \times 0 + 3b = 15 \Rightarrow b = 5/3$ (this is not possible since b has to be an integer value)

Hence, the house has parking space $\Rightarrow 12 + 5 \times 0 + 3b = 15 \Rightarrow b = 1 \Rightarrow$ One neighbor house is occupied among D1 and F1.

Let's take the case of house D1 being occupied and F1 being empty. In that case, the value of house F1 would be $10(\text{there is no parking space}) + (5 \times 0) + (3 \times \text{the number of neighbours})$

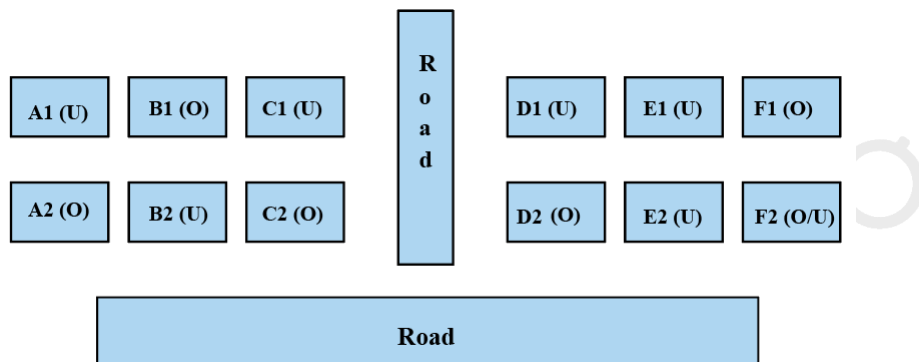
Here, even if we take the number of neighbors to be 1, which is the maximum for F1 in this case, the value of F1 would be a maximum of 13. This is lower than the lowest-value house in block YY. Therefore, F1 cannot be empty.

Since F1 is occupied and we know that there is only one house occupied in row 1 of each block, D1 becomes unoccupied. D2 becomes occupied because it is given in the question that each of Column-D and Column-F has at least one occupied house.

Here, the value of D1 is 18 as D2 is occupied.

We do not know the status of house F2.

Therefore, the final diagram is given below:



30.3

From the diagram, we can see that 3 houses are vacant in block XX.

[VIDEO SOLUTION](#)



31.C

From the diagram, we can see that B1 and D2 are definitely occupied. The rest of the options are not definitely correct.

The correct options are both B and C.

[VIDEO SOLUTION](#)

32.D

From the diagram, we can say that the number of vacant houses in Row 2 in Block XX is 1, and the number of vacant houses in Row 2 in Block YY is either 1 or 2.

Hence, the total number of vacant houses is either 2 or 3

The correct option is D

[VIDEO SOLUTION](#)

33.21

From the diagram, the vacant house with the maximum possible quoted price in column E is E2 when F2 is occupied.

The maximum possible quoted price of E2 is $10 + 5 \times 1 + 3 \times 2 = 21$ Lacs. (E2 has no parking space because E1 has the parking space, and it is given that there is only one house with parking space in Block YY.)

[VIDEO SOLUTION](#)

34.A

From the diagram, we can see that E1 has the parking space (case 2).

The correct option is A

[VIDEO SOLUTION](#)

Explanation [35 - 39]:

Given that the means of the ratings given by R1, R2, R3, R4 and R5 were 3.4, 2.2, 3.8, 2.8 and 3.4 respectively.

=> The sum of ratings given by R1, R2, R3, R4, R5 are $5 \times \text{means} = 17, 11, 19, 14, \text{ and } 17$ respectively.

Similarly the sum of ratings received by U, V, W, X and Y are $5 \times \text{means} = 11, 19, 17, 18, \text{ and } 13$ respectively.

Also capturing the absolute data given in the partial information (a) and (b) and representing as a table, we get:

	R1	R2	R3	R4	R5	Total
U	1					11
V					5	19
W	5	1	5			17
X		5	5			18
Y		1	1			13
Total	17	11	19	14	17	

Now,

Consider U

Given median = 2, mode = 2 and range = 3

=> His ratings should be of the form 1, a, 2, b, 4 => $1 + 2 + 4 + a + b = 11$ => $a + b = 4$. For mode = 2 => $a = b = 2$

=> U's ratings are 1, 2, 2, 2, 4.

Consider V

Given median = 4, mode = 4 and range = 3

=> His ratings should be of the form 2, a, 4, b, 5 => $2 + 4 + 5 + a + b = 19$ => $a + b = 8$ => For mode = 4 => $a = b = 4$

=> V's ratings are 2, 4, 4, 4, 5.

Consider W

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $1 + a + 4 + 5 + 5 = 17$ => $a = 2$

=> W's ratings are 1, 2, 4, 5, 5.

Consider X

Given median = 4, mode = 5 and range = 4

=> His ratings should be of the form 1, a, 4, 5, 5 => $a + 1 + 4 + 5 + 5 = 18$ => $a = 3$

=> X's ratings are 1, 3, 4, 5, 5

Consider Y

Given median = 3, mode = 1 & 4, Range = 3

=> His ratings are 1, 1, 3, 4, 4.

Capturing this data in the table, we get:

	R1	R2	R3	R4	R5	Total	Entries
U	1					11	2,2,2,4
V					5	19	2,4,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R3 => The two missing entries should add up to $19 - 1 - 5 - 5 = 8$, (only possibility is 4 + 4) => We can fill the row "U" and 4 in the row "V"

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V			4		5	19	2,4,4
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Now, consider column R2 => Missing entry should be $11 - 2 - 1 - 5 - 1 = 2$

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5			17	2,4
X		5	5			18	1,3,4
Y		1	1			13	3,4,4
Total	17	11	19	14	17		

Consider column R1, the missing elements should add up to $17 - 5 - 4 - 1 = 7$ (3 + 4 or 4 + 3) ----(1)

Consider R5, the missing elements should add up to 10 => $2 + 4 + 4$ or $4 + 3 + 3$ (not possible) as (1) requires a 3.

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X		5	5		4	18	1,3
Y		1	1		4	13	3,4
Total	17	11	19	14	17		

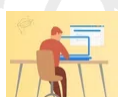
Now, we can fill column R1 as 3 + 4 and the remaining in column R4 and we can get the complete table

	R1	R2	R3	R4	R5	Total	Missing Entries
U	1	2	4	2	2	11	
V	4	2	4	4	5	19	
W	5	1	5	4	2	17	
X	3	5	5	1	4	18	
Y	4	1	1	3	4	13	
Total	17	11	19	14	17		

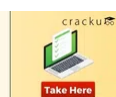
35.0

=> All ratings can be determined uniquely => 0.

VIDEO SOLUTION



CAT Sectional Tests



36.0

R2 gave ratings of 1, 1, 2, 2, 5 => He gave 4 to 0 workers => 0 is the answer.

VIDEO SOLUTION

37.3

=> From the table, we can see that R1 gave a rating of 3 to Xavier.

VIDEO SOLUTION

38.4

=> Ratings give by R3 are 1, 4, 4, 5, 5 => Median = 4.

VIDEO SOLUTION

39.C

=> R2 median rating is 2 => given to 2 workers

=> R5 median rating is 4 => given to 2 workers

=> R4 median rating is 3 => given to only 1 worker.

=> R3 median rating is 4 => given to 2 workers.

VIDEO SOLUTION

Explanation [40 - 44]:

Now, we know there is only 1 candidate from OQ, which means that the number of non-candidate voters in OQ will be 4.

We also know that the non-candidates in a particular department voted as a block, and we also know that the least number of non-candidate voters in a particular department can be 1 (BH, 3-2 faculty).

Now, we also know that R got 5 votes from non-candidates.

Now we can write 5 as

i)5

ii)4+1

iii)3+2

Considering case (i) 4+1. This is only possible when there is 1 candidate from OQ, and there are 2 candidates from BH. This implies that the number of candidates in FA and MQ is 1. Now, if we consider FA and MQ and put only 1 candidate there, it implies that there are 15 non-candidate voters between them. Now we know this is not possible since the maximum number of non-candidate voters a candidate can get is 13. (Please note that non-candidates of a particular department vote as a block).

On similar grounds, we can eliminate Case (iii) as it also implies there is only 1 candidate in FA and MQ.

Now, considering Case (i), we know that 5+0 will happen only one when there are 5 non-candidates in a single department. This is only possible in MS (Out of 7, there will be 2 candidates and 5 non-candidates).

So we can conclude that MS has 2 candidates and that they voted Prof. R.....(i)

We also know that Prof P got 2 votes from Non-candidates. This is only possible when BH has 1 candidate.

So, we can conclude that the number of professors in FA, MS, OQ, BH is 0,2,1,1

	FA	MS	OQ	BH
Total number of voters	9	7	5	3
Number of candidates	0	2	1	1
Number of non-candidates	9	5	4	2

Thus, we get the following table:

	P	Q	R	S
Total Votes	3	14	6	1
Candidate Vote	1(S)	1(R)	1(P)	1(Q)
Votes from non candidates	2	13	5	0
	BH	FA+OQ	MS	

Now, if we consider Department MS, we know that there are 2 candidates from MS and R can't be one of them as the people in that department voted for him..... (3 rd condition).

So the possible combinations of candidates in MS are (P,Q), (Q,S), (P,S).

Now we also know that no one can vote for a candidate in their own department, so we can eliminate (P,S) and (Q,S) as we know that S voted for P and Q voted for S).

So we can infer that P and Q are from MS.

	FA	MS	OQ	BH
Case 1	0	P, Q	R	S
Case 2	0	P, Q	S	R

40.A

Now, among the given options, Only Option A is true. Therefore, Option A is the correct answer.

VIDEO SOLUTION

41. **D**

Now, we can see that the number of votes that Prof Qureshi received from a single department can be 9 or 5 (if R is from OQ) or 4 (if R is not from OQ). So, among the options, only Option D can be true. Therefore, Option D is the correct answer.

[VIDEO SOLUTION](#)

42. **D**

We have been told Prof Samuel belongs to B&H. So we have to consider only Case 1. In Case 1 we can see that Prof Prakash belongs to MS and Prof Ramaswamy belongs to OQ.

Therefore, both the statements are true.

So, the correct answer is Option D

[VIDEO SOLUTION](#)

43. **B**

From the 2 cases, we can see that the candidate from OQ can either be Prof Ramaswamy or Prof Samuel.

Therefore, the correct answer is Option B.

[VIDEO SOLUTION](#)

44. **B**

Since Prof Qureshi belongs to MS, non-candidates from MS can't vote for him. We can see that the non-candidates from FA voted for him. So, only statement B is true. Therefore, the correct answer is Option B.

[VIDEO SOLUTION](#)

Quant

45. **B**

Prime Factorising 1134, we get $1134 = 2 \times 3^4 \times 7$ and $168 = 2^3 \times 3 \times 7$

1134^n is a factor of 168 $\Rightarrow$ the factor of 2 should be atleast 3, for 168 to be a factor $\Rightarrow n = 3$.

Now, $1134^n = 1134^3 = 2^3 \times 3^{12} \times 7^3$ is a factor of $168^m = (2^3 \times 3 \times 7)^m \Rightarrow m = 12$ as power of 3 should be atleast 12.

$\Rightarrow$ So, $m + n = 15$.

[VIDEO SOLUTION](#)



46. **C**

Given, $\log_x(x^2 + 12) = 4$

$$\Rightarrow x^2 + 12 = x^4$$

$$\Rightarrow x^4 - x^2 - 12 = 0$$

$$\Rightarrow x^4 - 4x^2 + 3x^2 - 12 = 0$$

$$\Rightarrow x^2(x^2 - 4) + 3(x^2 - 4) = 0$$

$$\Rightarrow (x^2 - 4)(x^2 + 3) = 0 \Rightarrow \text{since, } x \text{ is a positive real number (given)} \Rightarrow x = 2.$$

Now, Given $3 \log_y x = 1$

$$\Rightarrow \log_y x = \frac{1}{3}$$

$$\Rightarrow x = y^{\frac{1}{3}}$$

$$\Rightarrow y = x^3 \Rightarrow y = 8.$$

$$\Rightarrow x + y = 2 + 8 = 10.$$

[VIDEO SOLUTION](#)

47. **C**

Given, $\sqrt{5x+9} + \sqrt{5x-9} = 3(2 + \sqrt{2})$

$$\Rightarrow \sqrt{5x+9} + \sqrt{5x-9} = 6 + 3\sqrt{2}$$

$$\Rightarrow \sqrt{5x+9} + \sqrt{5x-9} = \sqrt{36} + \sqrt{18}$$

Comparing the L.H.S. and R.H.S.

$$\Rightarrow 5x + 9 = 36 \Rightarrow 5x = 27 \Rightarrow x = \frac{27}{5} \text{ (can be verified using the second term as well).}$$

$$\Rightarrow \sqrt{10x+9} = \sqrt{\left(10 \times \frac{27}{5}\right) + 9} = \sqrt{63} = 3\sqrt{7}$$

VIDEO SOLUTION

48. B

Given, $x^2 + (x - 2y - 1)^2 = -4y(x + y)$

$$\Rightarrow x^2 + 4xy + 4y^2 + (x - 2y - 1)^2 = 0$$

$$\Rightarrow (x + 2y)^2 + (x - 2y - 1)^2 = 0$$

For the L.H.S. of the equation to be 0, each of the square terms should be 0 (as squares cannot be negative)

$$\Rightarrow x - 2y - 1 = 0 \Rightarrow x - 2y = 1$$

VIDEO SOLUTION

49. 3

Let us consider 3 cases:

1) $x = 0$, This is a solution, as both L.H.S and R.H.S will be equal (0) when $x = 0$. (1 solution)

2) $x > 0$

$$\Rightarrow 2x(x^2 + 1) = 5x^2$$

$$\Rightarrow 2(x^2 + 1) = 5x$$

$$\Rightarrow 2x^2 - 5x + 2 = 0 \Rightarrow 2x^2 - 4x - x + 2 = 0$$

$$\Rightarrow 2x(x - 2) - 1(x - 2) = 0$$

$$\Rightarrow (x - 2)(2x - 1) = 0 \Rightarrow x = 2 \text{ or } 1/2 \Rightarrow (1 \text{ integer solution})$$

3) $x < 0$

$$\Rightarrow -2x(x^2 + 1) = 5x^2$$

$$\Rightarrow 2x^2 + 5x + 2 = 0$$

$$\Rightarrow 2x^2 + 4x + x + 2 = 0$$

$$\Rightarrow 2x(x + 2) + 1(x + 2) = 0$$

$$\Rightarrow (x + 2)(2x + 1) = 0 \Rightarrow x = -2 \text{ or } -1/2 \Rightarrow (1 \text{ integer solution})$$

So, the total number of integer solutions are 0, 2, -2 $\Rightarrow$ 3.

VIDEO SOLUTION

50. 2

Given that -2 is a root of the given cubic equation.

$\Rightarrow$ Dividing the given equation by $(x + 2)$, Using the Horner's method of synthetic division:

coefficient of x^2 is 1, and coefficient of x is $(2r+1)-2 = 2r-1$ and the constant term = $(4r-1)-2(2r-1) = 1$.

$\Rightarrow$ The quadratic obtained by dividing the cubic = $x^2 + (2r - 1)x + 1 = 0$, Since, this equation has 2 real roots $\Rightarrow$ Discriminant should be greater than 0

$$\Rightarrow (2r - 1)^2 > 4 \Rightarrow 2r - 1 > 2 \text{ or } 2r - 1 < -2 \Rightarrow r > 3/2 \text{ or } r < -1/2.$$

$\Rightarrow$ Minimum possible non-negative integer value of r is 2.

VIDEO SOLUTION

100+ CAT Quant Questions



51. 6

Given a and b are the distinct roots of the equation $2x^2 - 6x + k = 0$

$$\Rightarrow a + b = -(-6/2) = 3 \text{ (Sum of the roots)}$$

$$\Rightarrow ab = k/2 \text{ (Product of the roots)}$$

Now, $(a+b)$ and ab are the roots of the quadratic equation $x^2 + px + p = 0$

$$\Rightarrow a + b + ab = -p \Rightarrow 3 + k/2 = -p \text{ ---(1)}$$

$$\Rightarrow (a + b)(ab) = p \Rightarrow 3(k/2) = p \text{ ---(2)}$$

$$3 + \frac{k}{2} = -\frac{3k}{2} \Rightarrow 2k = -3 \Rightarrow k = -\frac{3}{2}$$

$$p = \frac{3k}{2} = \frac{3}{2} \left(-\frac{3}{2} \right) = -\frac{9}{4}$$

$$\Rightarrow 8(k-p) = 8 \left(-\frac{3}{2} + \frac{9}{4} \right) = -12 + 18 = 6$$

VIDEO SOLUTION

52. C

Given that in the final mixture, the ratio of coffee and cocoa is 16:9

Let us assume coffee is 16 units and cocoa is 9 units.

=> Initially, there are 25 units of coffee and 0 units of cocoa

Let's say x units of the mixture is removed and replaced with cocoa

=> Now, we have (25-x) coffee and x units of cocoa. => Mixture P

Now, if x units of the mixture is removed:

$$\text{Amount of coffee present} = (25-x) - \frac{(25-x)}{25} \times x$$

$$\Rightarrow (25-x) \left(1 - \frac{x}{25}\right) = 16$$

$$\Rightarrow (25-x)^2 = 16 \times 25$$

$$\Rightarrow 25-x = 20 \Rightarrow x = 5.$$

In mixture P, cocoa = x = 5

In mixture Q, cocoa = 9 units.

=> Required ratio = 5:9

[VIDEO SOLUTION](#)

53. D

The given time is 8:48 AM.

Angle made by hours hand w.r.t 12 is 8×30 (30 degrees in 1 hour) + 0.5×48 (0.5 degree in 1 minute) = $240 + 24 = 264$ degrees.

Angle made by minutes hands w.r.t 12 is $48 \times 6 = 288$ degrees.

=> The angle between them is $288 - 264 = 24$ degrees.

This should further increase by 12 degrees (50% of 24)

$$\text{After } m \text{ minutes, the further increase in angle} = (6 - 0.5)m = \frac{11}{2}m = 12 \Rightarrow m = \frac{24}{11}$$

[VIDEO SOLUTION](#)

54. C

Let us assume the initial selling prices of A and B is p.

Given, she made profit of 20% on A => $1.2 \times c = p \Rightarrow c = 5p/6 \Rightarrow$ cost of A is $\frac{5}{6}p$

Given, she made a loss of 10% on B => $0.9 \times c = p \Rightarrow c = 10p/9 \Rightarrow$ cost of B is $\frac{10}{9}p$

Now, she sold them at a price such that a 10% profit is made on B

$$\Rightarrow \text{Selling price} = s = 11/10 \times 10/9 p \Rightarrow \frac{11}{9}p$$

$$\Rightarrow \text{Profit \% on A} = \frac{\left(\frac{11}{9} - \frac{5}{6}\right)}{\left(\frac{5}{6}\right)} \times 100 = 46.66\% = \text{nearly } 47\%$$

[VIDEO SOLUTION](#)

55. C

Given the total number of kilometres travelled, including the trip = is 26862 Km, and the duration of the trip is 8 hrs.

If avg. speed of the car during the trip is 's' => the km travelled till just before the trip is $26862 - 8s$, which should also be a palindrome.

=> From the options if $s = 110 \Rightarrow$ The reading will be $26862 - 110 \times 8 = 25982$ (Not a palindrome)

=> If $s = 100 \Rightarrow$ The reading will be $26862 - 100 \times 8 = 26062 \Rightarrow$ It is a palindrome.

=> $s = 100$ is the correct option.

[VIDEO SOLUTION](#)

100+ CAT DILR Questions



56. A

Given that the average marks of 4 girls and 6 boys is 24.

Let us assume 'b' is the marks scored by a boy and 'g' is the marks scored by a girl.

$$\Rightarrow 4g + 6b = 10 \times 24 = 240 \text{ ---(1)}$$

Given that, $b \leq g \leq 2b$

We need to find the distinct possible values of $2g + 6b = 2g + 240 - 4g = 240 - 2g$.

From (1)

$$\text{when } b = g \Rightarrow 10g = 240 \Rightarrow g = 24$$

$$\text{when } b = g/2 \Rightarrow 7g = 240 \Rightarrow g = 240/7$$

=> $240 - 2g$ varies from $240 - 2 \times 24$ to $240 - 2 \times 240/7$

=> 171.42 to 192 => Integer values of 172 to 192 => 21 values.

VIDEO SOLUTION

57. C

Given, the salaries of Sita, Gita and Mita are initially in the ratio 5 : 6 : 7, respectively, Let us assume their salaries are 5p, 6p and 7p.

They get salary hikes of 20%, 25% and 20%, respectively.

=> Their salaries are $6/5 \times 5p$, $5/4 \times 6p$ and $6/5 \times 7p$ => 6p, 7.5p, 8.4p

Now, Sita and Mita get salary hikes of 40% and 25%, respectively

=> Sita's salary = $1.4 \times 6p = 8.4p$ and Mita's salary = $1.25 \times 8.4p = 10.5p$

Let Gita's salary be 'g' after hike

=> $3g = 8.4p + g + 10.5p$ => $2g = 18.9p$ => $g = 9.45p$

=> Hike % = $\frac{(9.45 - 7.5)}{7.5} \times 100 = 26\%$

VIDEO SOLUTION

58. 972

Let us assume the speeds of Arvind and Surbhi are 'a' and 's', respectively.

Let us say they meet after 't' hours

=> Arvind travelled $s \times t$ distance in 6 hrs and Surbhi travelled $a \times t$ in 24 hrs

=> $s \times t = a \times 6$ and $a \times t = s \times 24$ => $t^2 = 6 \times 24$ => $t = 12$

Given $a = 54$ => $s \times 12 = 54 \times 6$ => $s = 27$.

=> Total distance between A and B is $(s+a) \times t = (54+27) \times 12 = 81 \times 12 = 972$ Kms.

VIDEO SOLUTION

59. 20808

Anil invested 22000 for 6 years at 4% interest compounded half-yearly

=> Amount = $22000 (1.02)^{12}$

Let Sunil invest 'S' rupees for 5 years at 4% C.I. half-yearly and 10% S.I. for 1 additional year

=> Amount = $S (1.02)^{10} (1.1)$

Given that the both amounts are equal

=> $22000 (1.02)^{12} = S (1.02)^{10} (1.1)$

=> $S = \frac{22000 (1.02)^2}{1.1} = 20808$

VIDEO SOLUTION

60. 27

Let us assume the efficiencies of Amal, Sunil, and Kamal are a, s, and k, respectively.

Given that they are in H.P.

=> $\frac{2}{s} = \frac{1}{a} + \frac{1}{k}$ ---(1)

Also, given that Kamal takes twice as much time as Amal to do the same amount of job

=> $a = 2k$

Given that when Amal and Sunil work for 4 days and 9 days, respectively, Kamal needs to work for 16 days to finish the remaining job.

=> If W is the total work => $4a + 9s + 16k = W$.

from (1) $\frac{2}{s} = \frac{1}{a} + \frac{1}{k}$ => $a = \frac{3}{2}s$ and $k = \frac{3}{4}s$

=> $4 \left(\frac{3s}{2} \right) + 9s + 16 \left(\frac{3s}{4} \right) = W$

=> $6s + 9s + 12s = W$

=> $27s = W$ => $s = \frac{W}{27}$

=> Sunil will take 27 days to finish the work when working alone.

VIDEO SOLUTION

100+ CAT VARC Questions



61. B

Given equation of circle = $x^2 + y^2 + 4x - 6y - 3 = 0$

Center of the circle is $(-2,3)$ and radius of the circle $= \sqrt{g^2 + f^2 - c} = \sqrt{4 + 9 + 3} = 4$

Let us assume the point of the intersection of the tangents is (h,k)

The angle made by the line joining (h,k) to the centre makes an angle of 30 degrees with the tangent, and $\sin(30)$ will be the ratio of the radius and the distance between the center and (h,k)

$$\Rightarrow \sin(30) = \frac{4}{\sqrt{(h+2)^2 + (k-3)^2}}$$

Squaring on both sides:

$$\frac{1}{4} = \frac{16}{(h+2)^2 + (k-3)^2}$$

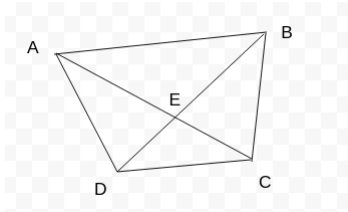
$$\Rightarrow (h+2)^2 + (k-3)^2 = 64$$

When $x = 6 \Rightarrow h = 6 \Rightarrow 64 + (k-3)^2 = 64 \Rightarrow k = 3$.

$\Rightarrow$ required point is $(6,3)$

[VIDEO SOLUTION](#)

62. C



Given ABCD is a cyclic quadrilateral.

Angle ADB = Angle ACB (Angle subtended by chord on the same side of arc)

Angle DAC = Angle DBC (Angle subtended by chord on the same side of arc)

$\Rightarrow$ Triangles AED and BEC are similar triangles

Similarly triangles AEB and DEC are also similar using AA similarity property.

Now, given that $AB : CD = 2 : 1$ and $BC : AD = 5 : 4$

$AE/BE = AD/BC = 4/5$ (Similar Triangles AED and BEC)

$BE/CE = AB/CD = 2/1$ (Similar Triangles AEB and DEC)

Multiplying both, we get $AE/CE = 8/5$.

[VIDEO SOLUTION](#)

63. 2

Given that ABC is a right-angled triangle with $AB = 5$ and $BC = 12 \Rightarrow$ Area of the triangle $= 0.5 * 5 * 12 = 30$.

Let us assume $BP = p$, $BQ = q$

$\Rightarrow$ Area of ABP $= 0.5 * 5 * p = 2.5p$

$\Rightarrow$ Area of ABQ $= 0.5 * 5 * q = 2.5q$

Given the area of ABC is 1.5 times that of ABP

$\Rightarrow 30 = 1.5 * 2.5p \Rightarrow 20 = 2.5p \Rightarrow p = 8$.

Given Areas of ABP, ABQ and ABC are in A.P. $\Rightarrow 2 * 2.5q = 2.5 * 8 + 30 \Rightarrow 5q = 50 \Rightarrow q = 10$.

$PQ = BQ - BP = q - p = 10 - 8 = 2$.

[VIDEO SOLUTION](#)

64. B

Given that $\frac{1}{\sqrt{y} + \sqrt{z}}$ is the arithmetic mean of $\frac{1}{\sqrt{x} + \sqrt{z}}$ and $\frac{1}{\sqrt{x} + \sqrt{y}}$

$$\Rightarrow \frac{2}{\sqrt{y} + \sqrt{z}} = \frac{1}{\sqrt{x} + \sqrt{z}} + \frac{1}{\sqrt{x} + \sqrt{y}}$$

$$\Rightarrow 2(\sqrt{x} + \sqrt{z})(\sqrt{x} + \sqrt{y}) = (\sqrt{y} + \sqrt{z})(\sqrt{x} + \sqrt{y} + \sqrt{x} + \sqrt{z})$$

$$\Rightarrow 2(x + \sqrt{xy} + \sqrt{xz} + \sqrt{yz}) = 2\sqrt{xy} + y + \sqrt{yz} + 2\sqrt{xz} + \sqrt{yz} + z$$

$$\Rightarrow 2x = y + z$$

$\Rightarrow y, x, z$ are in A.P. as x is the arithmetic mean of y and z .

[VIDEO SOLUTION](#)

65. C

1-digit numbers $\Rightarrow$ We have 1 to 9 $\Rightarrow 9$

2-digit numbers $\Rightarrow x, y$, we have 9 ways to choose x from 1 to 9 $\Rightarrow 9$ ways and 9 ways to choose y (0 to 9 except x) $\Rightarrow 9 * 9 = 81$

3-digit numbers $\Rightarrow x, y, z$, we have 9 ways to choose x , 9 ways to choose y and 8 ways to choose $z \Rightarrow 9 * 9 * 8 = 648$.

Total numbers till 1000 without digits repeated in them is $9 + 81 + 648 = 738$.

 VIDEO SOLUTION

CAT Expected Questions PDF



66. 19

Given on day-1, there are 2 organisms.

On day-2, there are $2 \cdot 2 + 3 = 7$ and on day-3, there are $2 \cdot 7 + 3 = 17$.

Let us try to form a pattern:

$$2 = 2 + 0 \quad (n = 1)$$

$$7 = 4 + 3 \quad (n = 2)$$

$$17 = 8 + 9 \quad [8 + 3 \cdot 3] \quad (n = 3)$$

$$37 = 16 + 21 \quad [16 + 3 \cdot 7] \quad n = 4$$

$$T(n) = 2^n + 3(2^{n-1} - 1)$$

We know that $2^{20} = 2^{10} \times 2^{10} = 1024 \times 1024$, which is more than 1 million.

Let us check for $n = 19$

$$2^{19} + 3(2^{18} - 1) = 2^{19} + 3 \cdot 2^{18} - 3 = 2 \cdot 2^{19} + 2^{18} - 3 = 2^{20} + 2^{18} - 3, \text{ which is more than 1 million.}$$

Let us check for $n = 18$

$$\Rightarrow 2^{18} + 3(2^{17} - 1) = 2^{18} + 3 \cdot 2^{17} - 3 = 2 \cdot 2^{18} + 2^{17} - 3 = 2^{19} + 2^{17} - 3 \text{ which is not more than a million.}$$

$\Rightarrow n = 19$.

 VIDEO SOLUTION